ENTERPRISE ENGINEERING ARCHITECTURE & PRODUCTION COCKPIT

Evolve AI Enterprise Delivery Studio

The Air-Gapped Engineering Cockpit for Forward Deployed AI — Built to solve the 4 critical failure modes of enterprise GenAI deployments.

🔒 01. Data Exfiltration & CISO Veto

100% AIR-GAPPED

Solved: 100% offline local inference (`Ollama`, `local Gemma`, `vLLM`) with zero outbound network calls. Credentials encrypted in native OS keyrings (Windows DPAPI, macOS Keychain).

⚖️ 02. Nondeterministic Silent Drift

<5MS • 0% DRIFT

Solved: Mathematical Rule-First Decision Gates evaluate latency & arithmetic, locking **Level 1 deterministic rules** over LLMs. Hybrid RAG (HNSW + BM25) delivers 100% SKU/domain recall.

💸 03. Runaway Token OpEx & GPU Sinkhole

HARDWARE SIZED

Solved: Automatically profiles host CPU/RAM/VRAM to fit local models before download; eliminates unpredictable per-token SaaS invoices and external GPU cluster lease overhead.

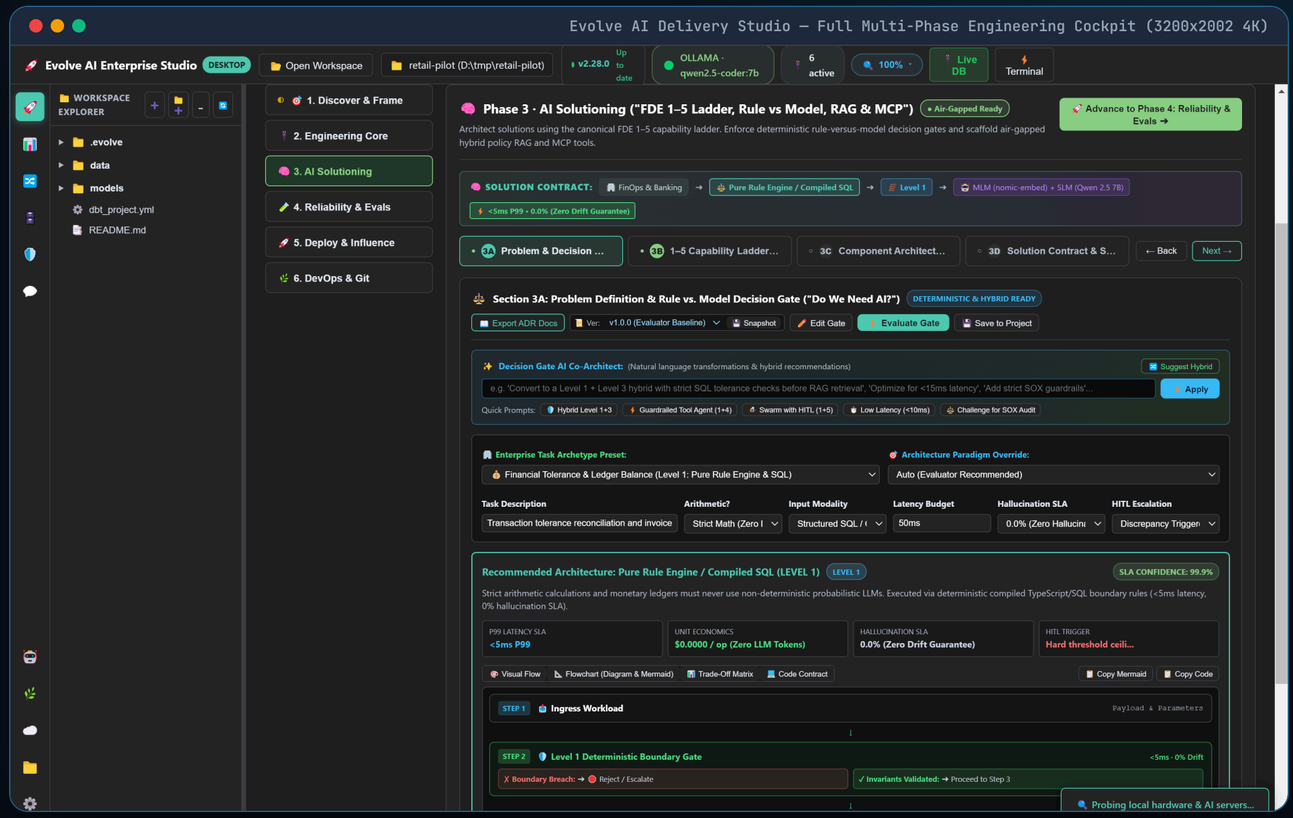
🍝 04. Spaghetti "Glue-Code" Technical Debt

4-WAY QUAD BUS

Solved: Multi-Agent Studio unifies App Bus, A2A Mesh, Vector DB, and Client DB with Oracle/DB2 2PC write-back, plus 1-click Terraform & Kubernetes production scaffolding.

PERIMETER: Air-Gapped On-Prem • Private VPC

INTEGRITY: Zero Guesswork • 100/100 Security Score



🌐 6-Phase Systematic Lifecycle

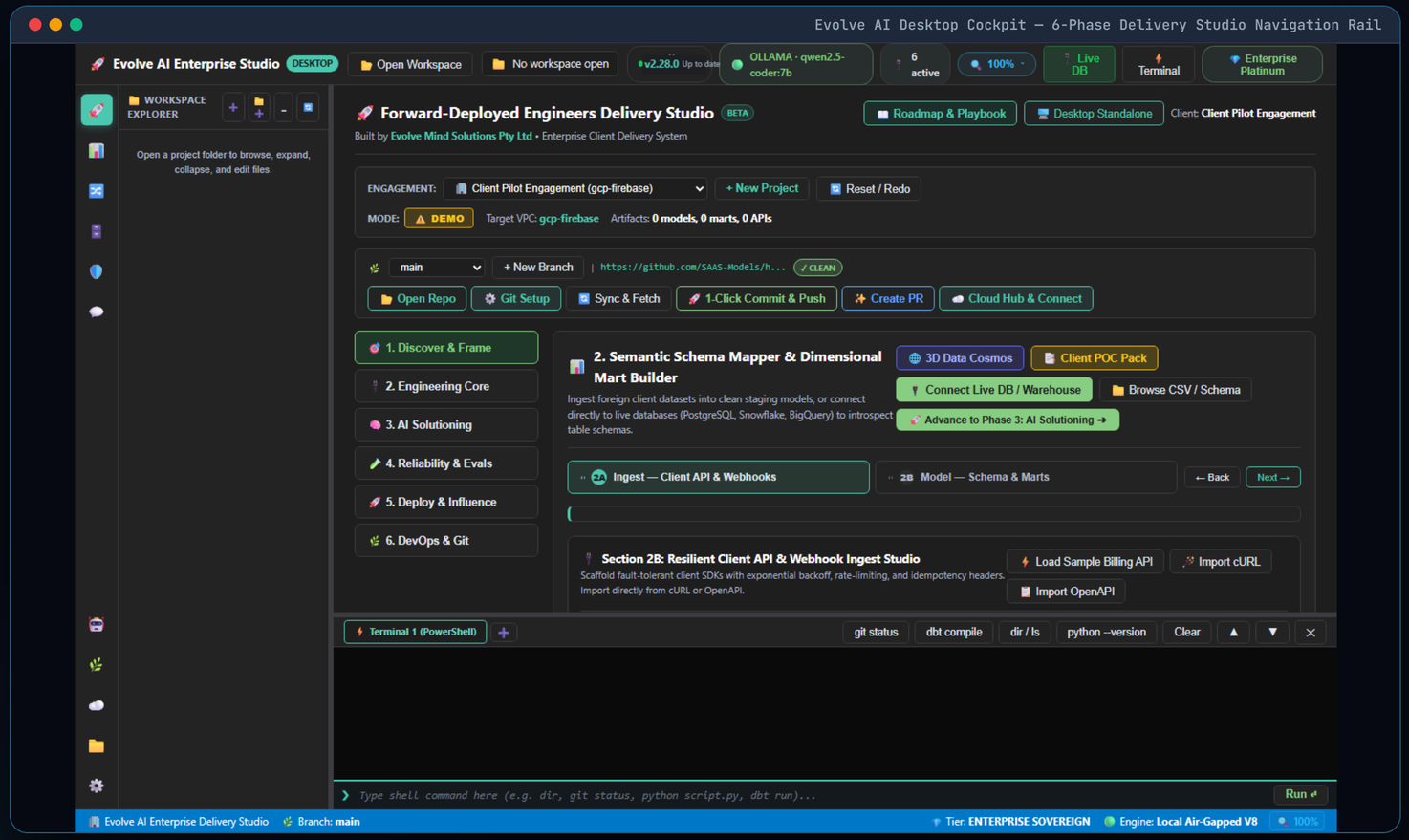
🤖 4-Way Multi-Agent Quad Bus

🌿 Dedicated Green Leaf Git Hub

AUDITABLE, REPEATABLE CLIENT ENGAGEMENT RUNBOOK

One Unified Studio Following the Whole Client Engagement

Move seamlessly from boardroom scoping to production-grade deployment within a single cockpit engineered specifically for Forward Deployed Engineering teams.



PHASE 1

Discover & Frame

Define objectives, ROI metrics, compliance bounds, and sign-off gates.

PHASE 2

Engineering Core

Connect live databases, map 2D/3D schemas, and scaffold dimensional marts.

PHASE 3

AI Solutioning

Rule-vs-model gate, 8 RAG patterns, and dual-runtime MCP server scaffolding.

PHASE 4

AI Engineering

2D visual DAG canvas, multi-agent swarms, and live in-studio simulator.

PHASE 5

Evals & Deploy

Empirical latency/token testing, 1-click Terraform, Docker & K8s manifests.

PHASE 6

DevOps & Git

Dedicated Green Leaf pane, 1-click Git wizards, live diffs & AI commit drafting.

Efficiency Multiplier: Eliminates fragmented handoffs, ensuring client requirements remain anchored to code and architecture.

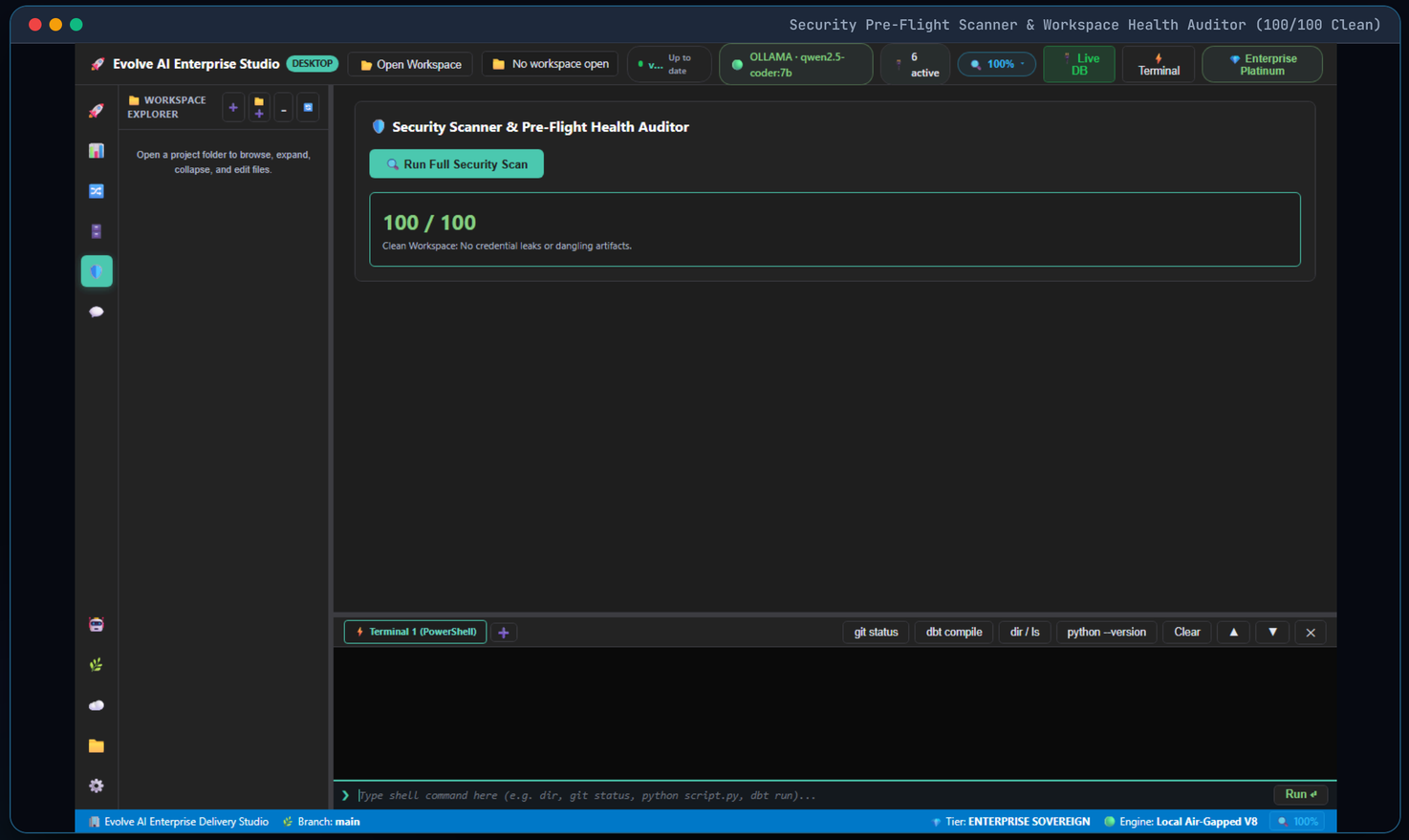
SECURITY PERIMETER • DATA SOVEREIGNTY

100/100 AUDIT SCORE

ZERO TRUST ARCHITECTURE & AIR-GAPPED HARDENING

Air-Gapped Local Execution & Zero Data Exfiltration Architecture

Engineered from the ground up for strictly regulated enterprise perimeters. All data, schemas, code, and model weights remain inside your firewall with verifiable zero-egress guarantees.



100% Air-Gapped Local Execution

ZERO EGRESS

Operates completely offline with zero outbound network calls. Compatible with local inference engines (`Ollama` , `local Gemma` , `ONNX` , `vLLM`) executing directly on workstation or on-prem cluster hardware.

Native OS Keyring Credential Vaults

ZERO PLAINTEXT

API tokens and database credentials are encrypted strictly inside native OS keyrings (Windows DPAPI, macOS Keychain, Linux Secret Service). Zero plaintext exposure in `.env` or configuration files.

Pre-Flight Security Scanner (100/100)

CISO AUDIT READY

Automated static analysis suite validates dynamic PII tokenization masks, audits dependency CVEs, flags hardcoded secret patterns, and enforces strict Ed25519 IPC licensing boundary checks.

CISO Approved: Passes enterprise information security audits on day one with zero external telemetry and verifiable air-gap guarantees.

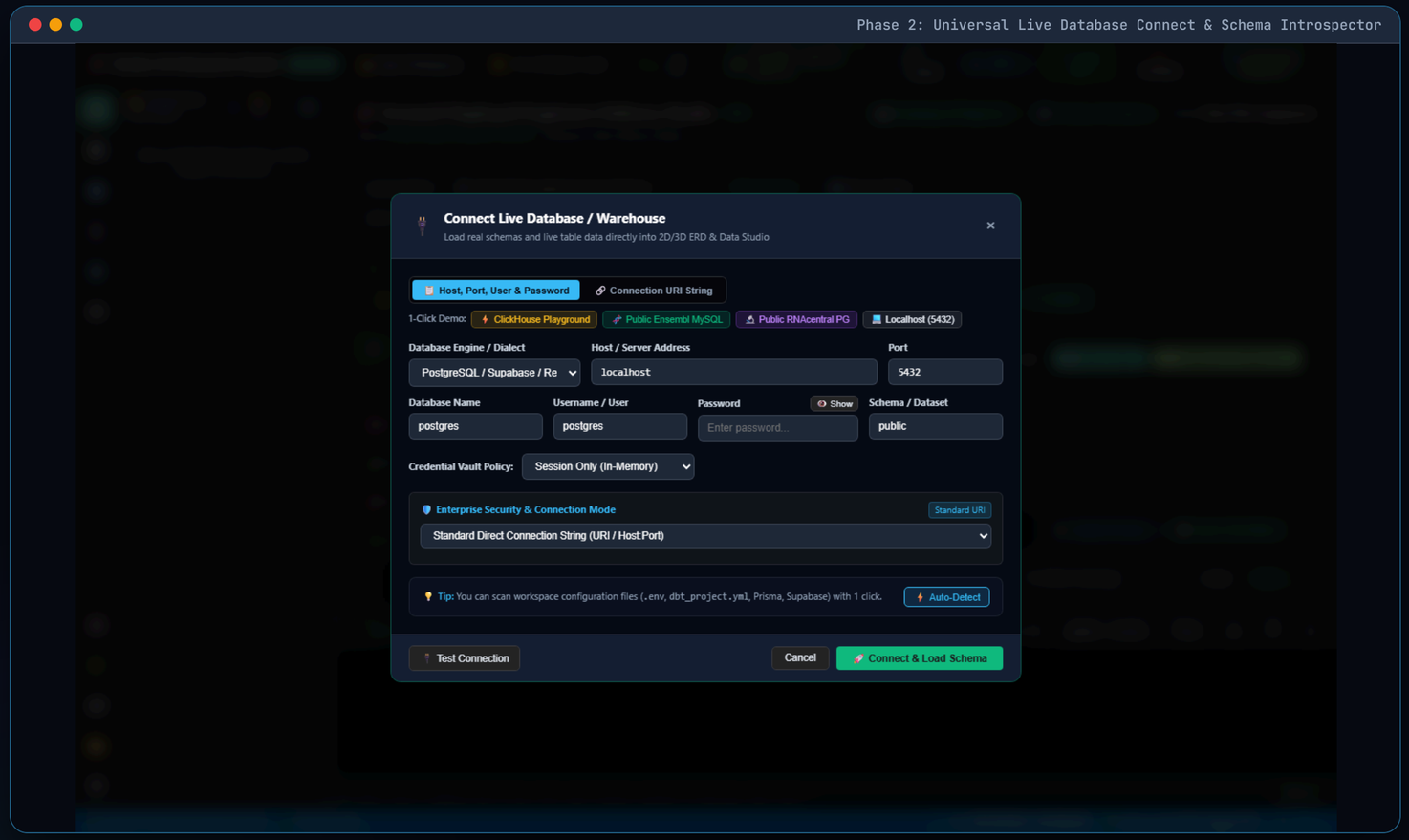
DATA INTROSPECTION • ZERO FOOTPRINT

SESSION-ONLY MEMORY

PHASE 2: UNIVERSAL LIVE WAREHOUSE INTROSPECTION MATRIX

Zero-Persistence Live Database Introspection & Governance

Harvest complex enterprise schemas, foreign key trees, and data profiles with ephemeral session-only memory and zero disk writes.



Universal Multi-Dialect Matrix

8+ WAREHOUSES

Native read drivers for Snowflake, BigQuery, Databricks, PostgreSQL, Oracle, IBM DB2, Microsoft SQL Server, and SQLite with automated metadata extraction.

Ephemeral Session-Only Memory

0 DISK WRITES

Connection pools and schema metadata exist strictly in volatile RAM. Never cached in dotfiles, settings, or unencrypted local storage.

Automated Star Schema & Mart Scaffolding

DBT • ANSI SQL

Discovers fact entities, dimension tables, and composite keys to automatically generate ANSI SQL and dbt staging models with full foreign key provenance.

Data Governance: Eliminate manual SQL dumps and insecure CSV extracts during client architectural discovery.

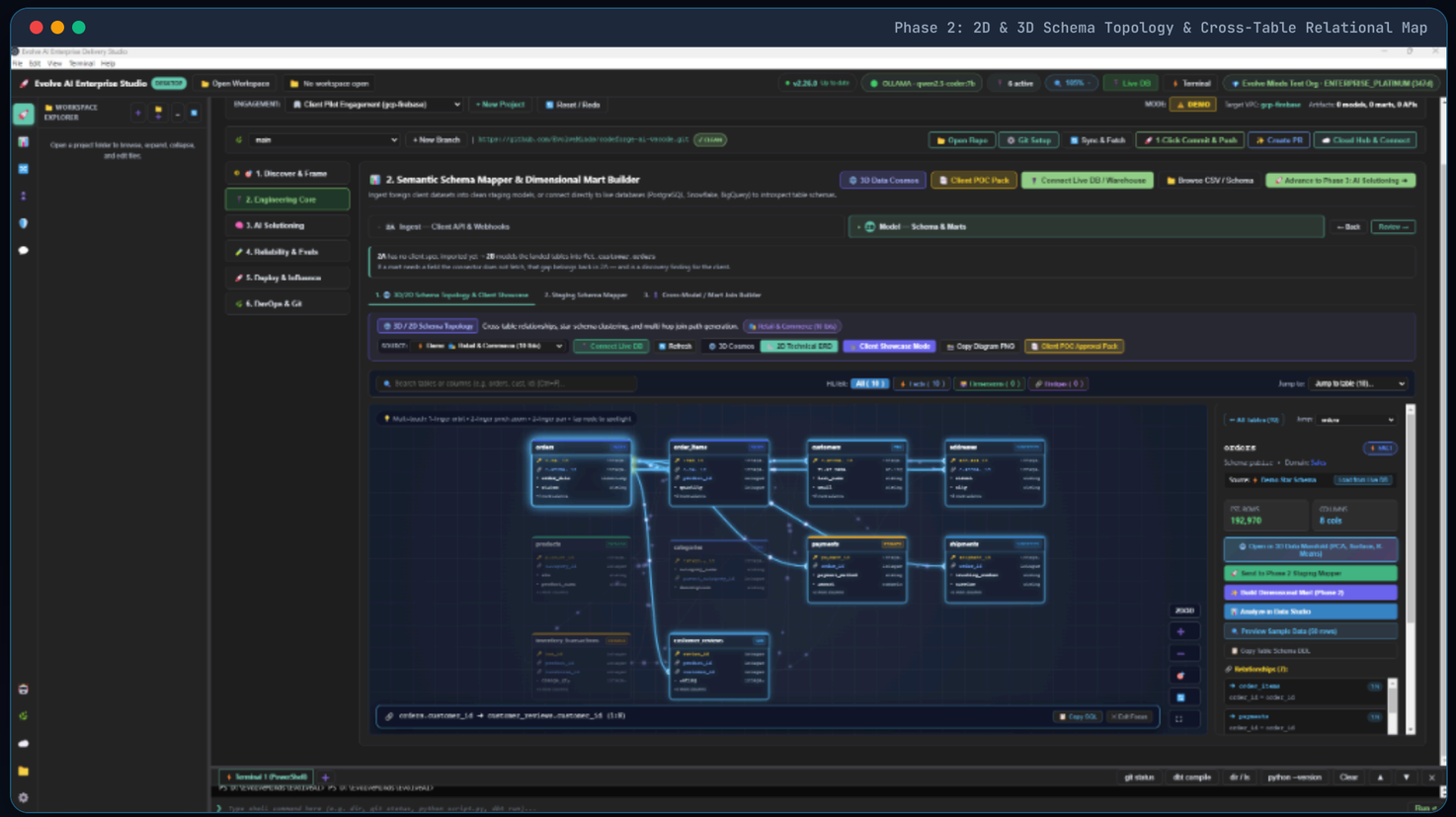
DATA TOPOLOGY • ARCHITECTURE VISUALIZATION

265+ TABLES AUTO-FIT

CROSS-TABLE TOPOLOGY & HIGH-SCALE MODELING

2D Technical ERD & 3D Celestial Topology Modeling

Eliminate schema ambiguity across massive enterprise data models (265+ tables) with column-level relational linking and join isolation.



Dual-Mode Topology Engine

2D / 3D DUAL

Switch between an interactive 20×14 technical ERD grid and a 3D Celestial star cluster with responsive auto-fit for 265+ enterprise tables.

Column-to-Column Illuminated Bezier Pinning

[PK] → [FK]

Illuminated cubic bezier curves connecting exact `[PK]` to `[FK]` socket pins with live join formula tooltips, multi-hop join resolution, and cardinality metrics.

Table Isolation Mode (4% Ghost Opacity)

4% GHOST OPACITY

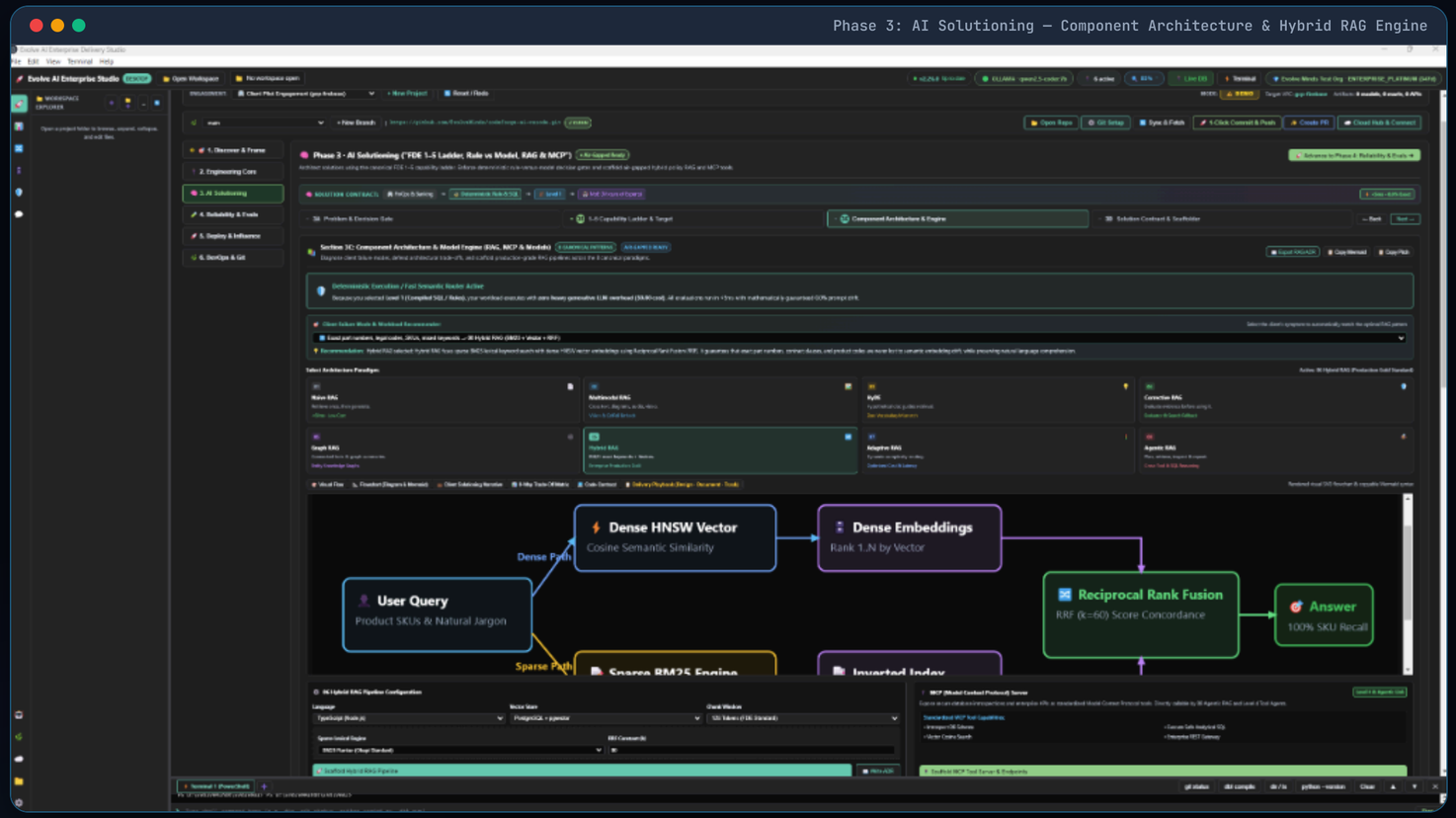
Click any connection line to dim unrelated tables to **4% blueprint ghost opacity**, spotlighting the transactional path with photon pulses and ANSI SQL generation.

Architecture Speed: Architects instantly understand multi-table join trees, partition keys, and relational dependencies without manual documentation.

PHASE 3: GOVERNANCE, CAPABILITY LADDER & HYBRID RETRIEVAL

Deterministic Decision Gates & The 5-Level Capability Ladder

Eliminate unpredictable prompt drift and runaway token OpEx. Enforce mathematical decision gates that prioritize compiled rule engines when probabilistic models are overkill.



⚖️ Rule vs. Model Decision Gate

`<5ms • 0% DRIFT`

Evaluates latency budgets, arithmetic determinism, and hallucination bounds. Automatically locks **Level 1 deterministic rules** over LLMs for ledger audits, compliance checks, and routing (`<5ms` , 0% drift).

🧠 Hybrid RAG Retrieval Engine

`100% SKU RECALL`

Production visual pipeline fusing **Dense HNSW Vectors** (cosine semantic match) and **Sparse BM25 Engines** via Reciprocal Rank Fusion (RRF), eliminating missing part numbers, SKU codes, and enterprise acronyms.

🛠️ Dual-Runtime MCP Tool Server

`NODE / PYTHON`

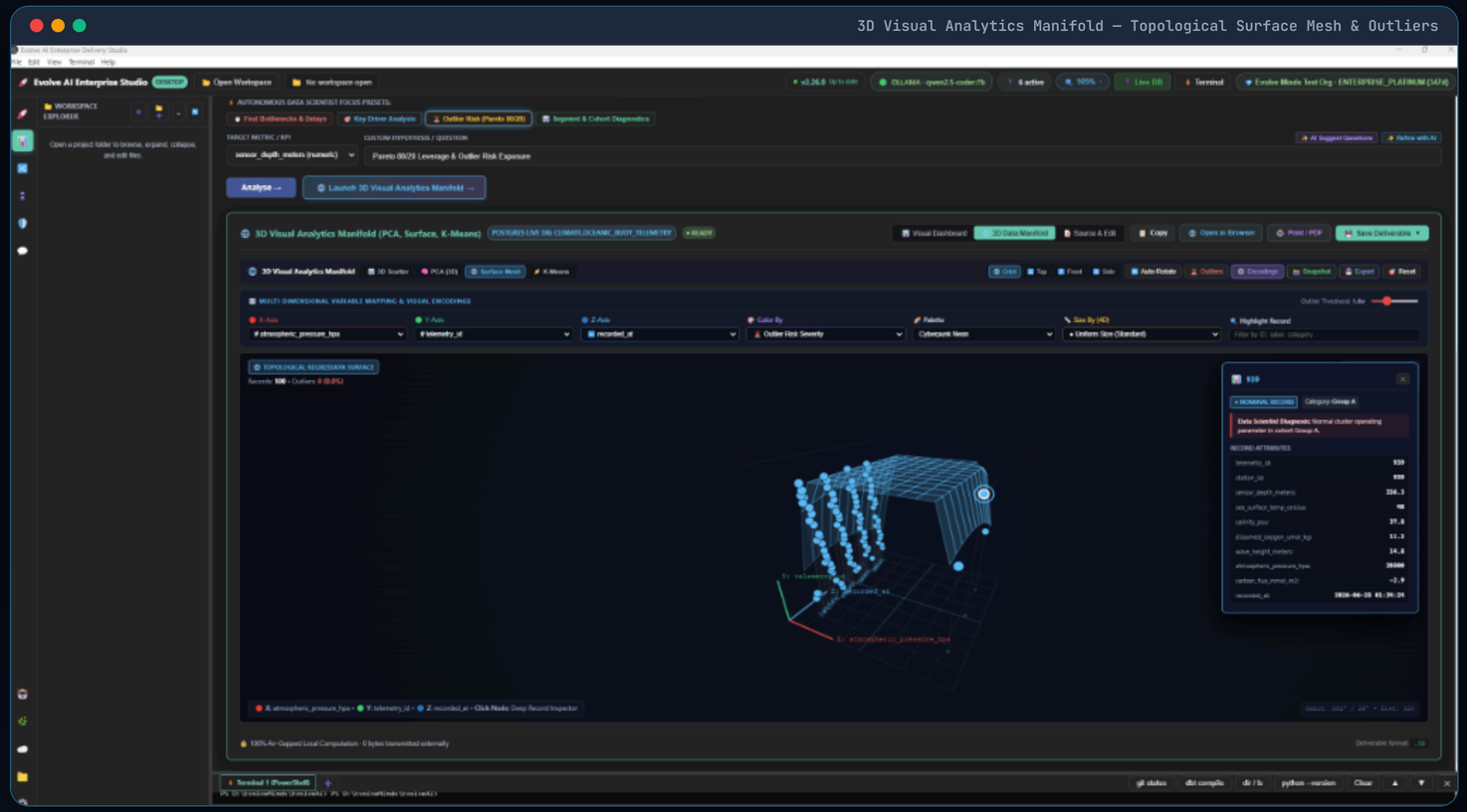
Auto-generates production TypeScript and Python Model Context Protocol (MCP) servers, tool schemas, Claude Desktop JSON export (`claude_desktop_config.json`), and signed Architecture Decision Records (ADRs).

Financial & Technical Sanity: Save 90%+ in token costs by deploying LLMs only where probabilistic reasoning is strictly required.

DATA ANALYSIS STUDIO & AIR-GAPPED MATHEMATICAL DIAGNOSTICS

3D Visual Analytics Manifold & Topological Spatial Diagnostics

Perform multi-variate 3D PCA, K-Means cluster partitioning, and continuous regression surface meshes with 100% local in-memory computation and zero data transmission.



Topological Surface Mesh & 3D PCA

IN-MEMORY SVD

Converts multi-column database tables into interactive 3D spatial manifolds (Principal Component Analysis, K-Means clustering, and continuous regression surface meshes) rendered in a Cyberpunk Neon palette.

Autonomous Outlier & Pareto Risk

80/20 LEVERAGE

Automated Focus Presets detect operational bottlenecks, key metric drivers, and Pareto outlier risk, color-encoding data points by distance to cluster centroid with automated risk scoring.

Floating Deep Record Inspector

ROW-LEVEL TRACE

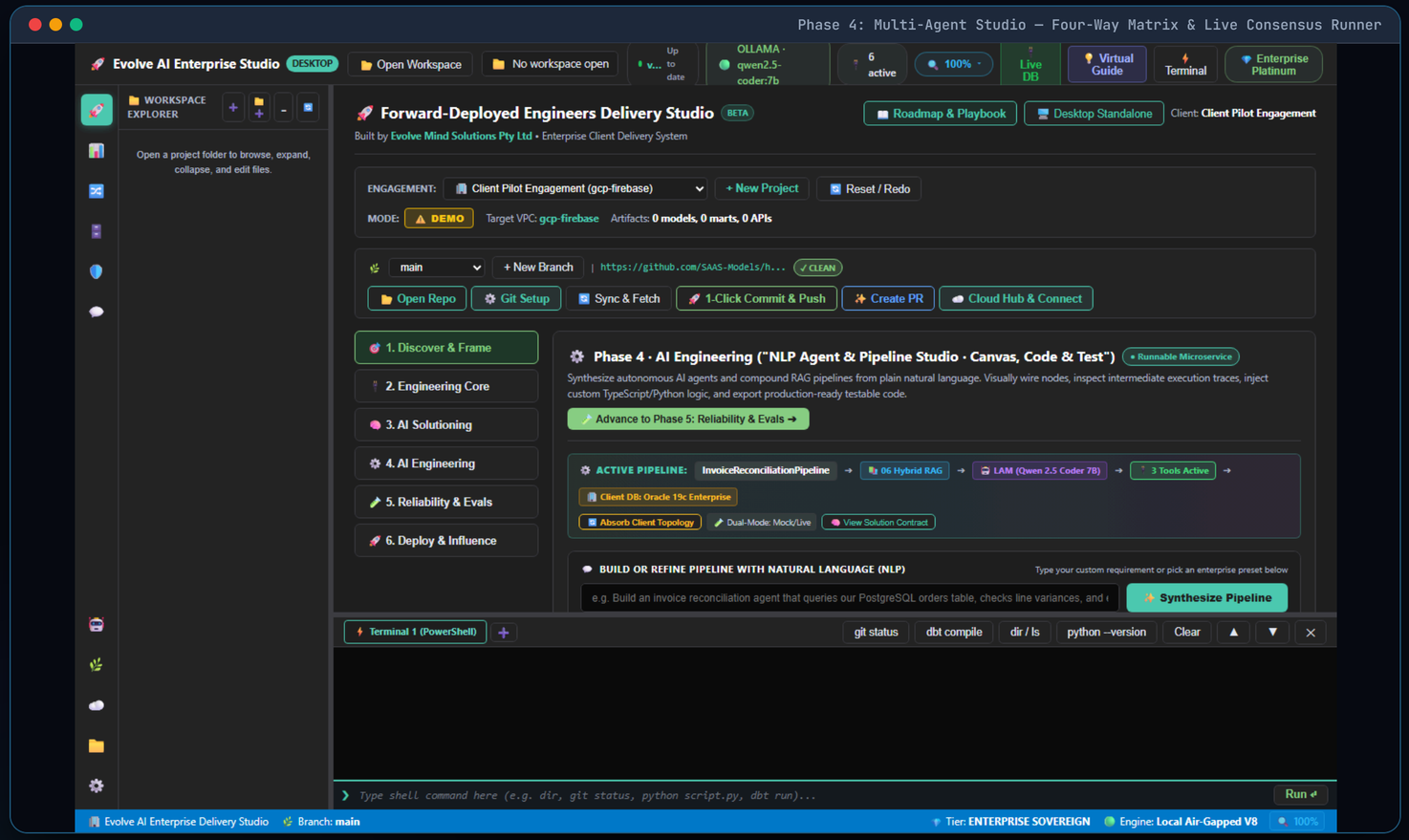
Interactive spatial drill-down: click any point in 3D space to inspect raw row attributes against cluster boundaries. All matrix factorizations compute locally in RAM with zero cloud egress.

Mathematical Rigor: All matrix factorizations and 3D geometry compute in local RAM with zero data leaving the host workstation.

PHASE 4: AGENTIC MESH ARCHITECTURE & LEGACY INTEGRATION

Multi-Agent Studio & Four-Way Communication Matrix

Architect, schedule, and test specialized autonomous AI agent swarms with direct legacy database connectivity and 2PC write-back.



Four-Way Quad-Bus Matrix

QUAD-BUS MESH

Bridges: **(1) Host App Bus** for lifecycle events, **(2) A2A Mesh** for peer consensus, **(3) RAG** for vector memory, and **(4) Client DB** for relational persistence.

Legacy DB Integration & 2PC Write-Back

ACID CONSISTENCY

Connects agents directly to legacy **Oracle, IBM DB2, and Teradata** warehouses with guaranteed Two-Phase Commit (2PC) ACID write-back and audit logs.

Drag-and-Drop Scheduler & Test Console

CODEGEN READY

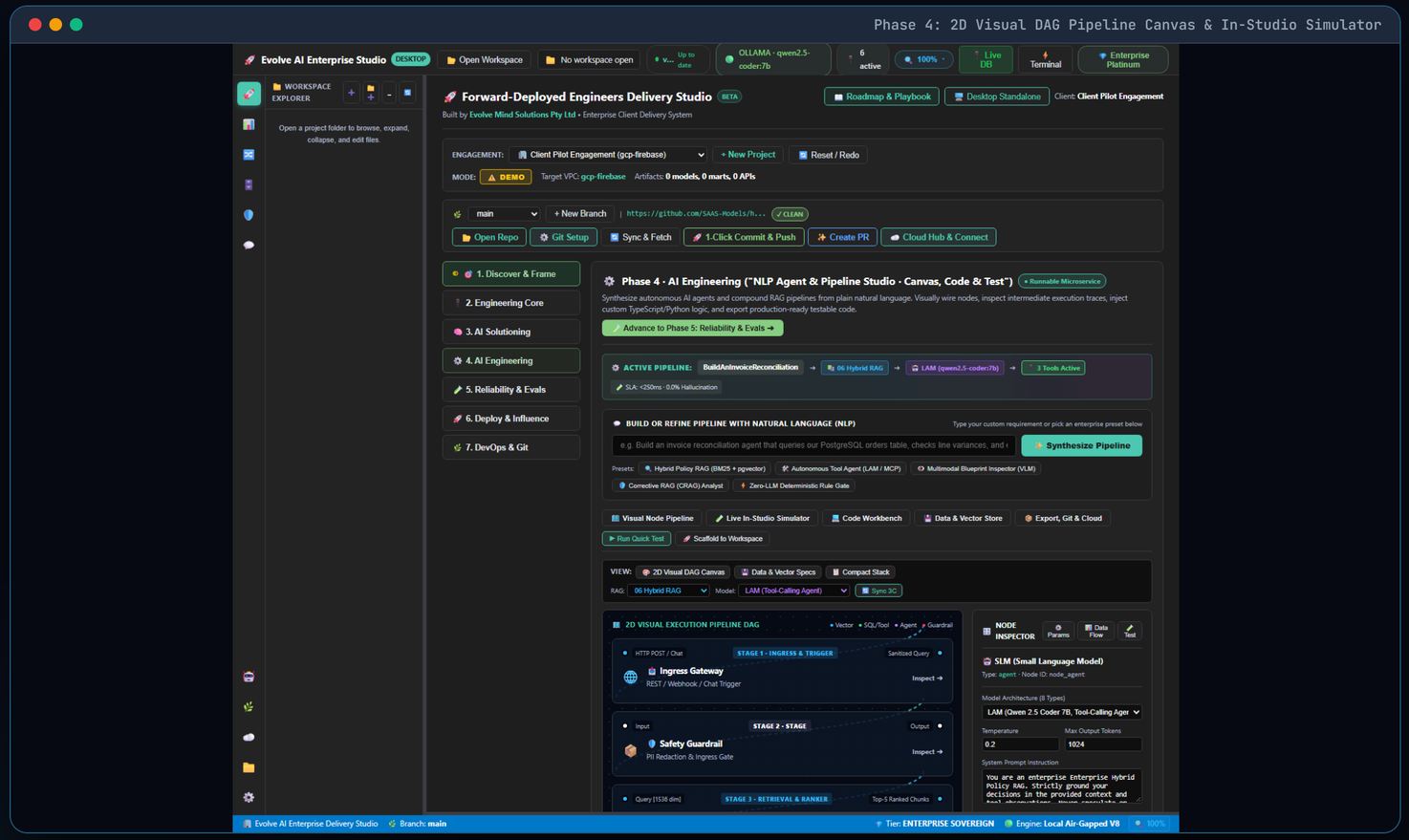
Priority queueing, concurrency limits, agent model ID selectors, and live consensus testing with automated TypeScript code generation (`multiAgentOrchestrator.ts`).

Production Multi-Agent: Replace toy single-turn prompts with deterministic, distributed agent swarms that coordinate safely.

PHASE 4: VISUAL DAG PIPELINE & VECTOR SPECIFICATIONS

2D Visual DAG Canvas & Pluggable Vector Store Architecture

Visually build, parameterize, and profile 8 canonical RAG architectures with multi-tier vector store targeting and dialect-agnostic SQL co-architecting.



Interactive 2D Visual DAG Pipeline

8 CANONICAL RAGS

Visually model and inspect Ingest, Embedding, Retrieval, Reranking, and Generation stages across 8 canonical RAG patterns (HyDE, Hybrid RRF, CRAG, Graph RAG, Self-RAG) with real-time parameter tuning.

Multi-Tier Vector Store Targeting

PLUGGABLE VECTORS

Target `pgvector`, `Pinecone`, `Qdrant`, or `Milvus` with 1-click presets. Fine-tunes HNSW/IVFFlat index parameters, embedding dimensions, and chunking strategy.

AI SQL Co-Architect & Dialect Transpiler

MULTI-DIALECT DDL

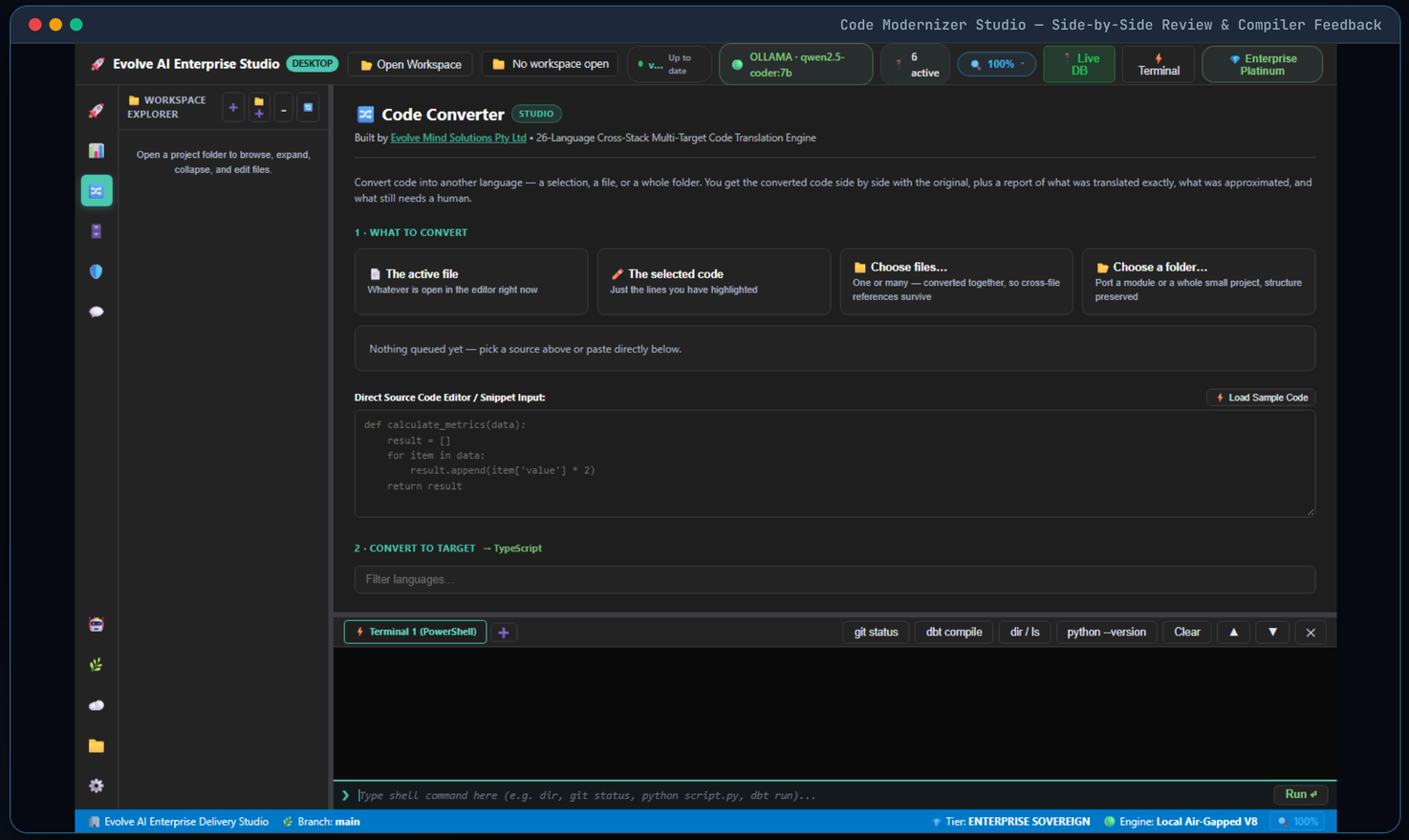
Convert, test, and validate complex queries across Snowflake, BigQuery, PostgreSQL, and Oracle with push-pull DDL editing, automated index recommendations, and explain-plan guidance.

Developer Ergonomics: Give engineers a visual workbench that turns abstract pipeline diagrams into deployable code instantly.

ENTERPRISE CODEBASE REFACTORING & MAINFRAME MIGRATION

26-Language Code Modernizer & AST-Grounded Transpiler

De-risk mission-critical legacy migrations by translating monolithic code into modern cloud-native targets with compiler verification.



26-Language Cross-Stack Translation

LEGACY → MODERN

Modernizes legacy codebases (COBOL, SAS, VBA, Fortran, Perl, PL/SQL) into cloud-native Python, TypeScript, Go, Rust, and dbt models with architectural parity.

Strict Side-by-Side Review Guarantee

ZERO BLIND OVERWRITES

Zero code is written to disk until engineers inspect side-by-side AST diffs, variable mappings, and logic parity indicators in the studio review pane.

Automated Syntax Self-Repair Loop

PARSER FEEDBACK

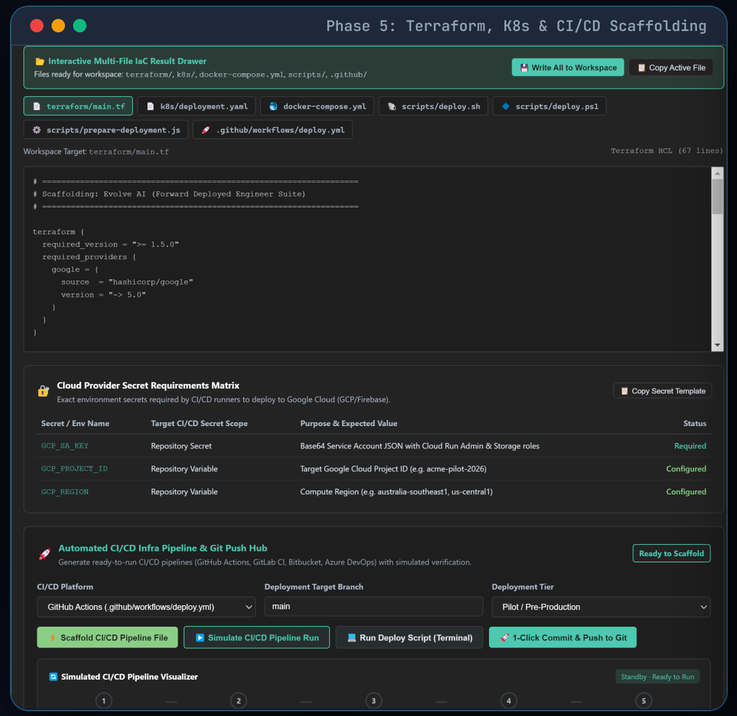
Validates generated code against local target language parsers. Syntactic failures are fed back for immediate self-correction before human review.

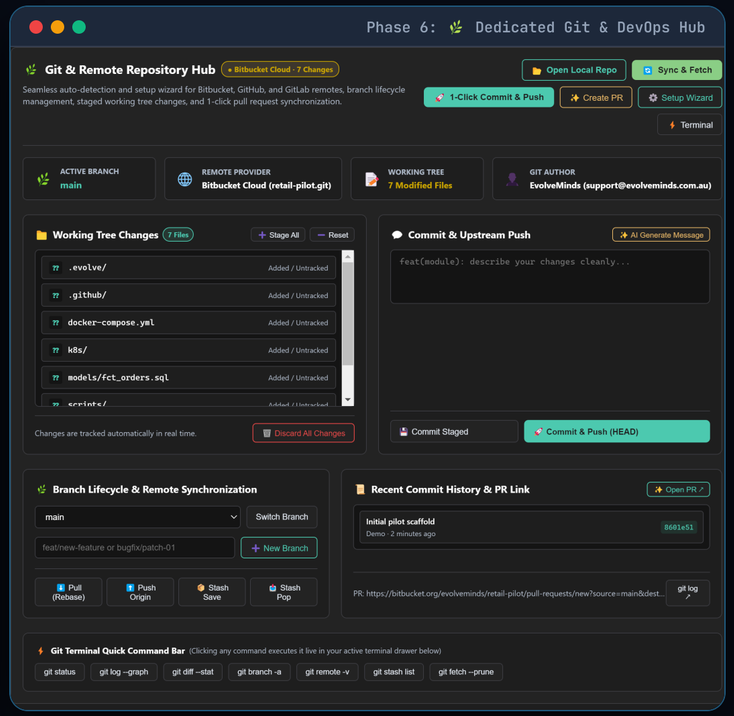
Modernization Velocity: Accelerate multi-year legacy migration projects into weeks without losing business logic or audit history.

PHASE 5 & 6: CLOUD SCAFFOLDING & DEDICATED GREEN LEAF GIT HUB

1-Click Multi-Cloud Scaffolding & Git DevOps Hub

Export production-ready Terraform, Docker, Kubernetes, and CI/CD pipelines, and manage repositories directly from the dedicated Green Leaf pane.





1-Click Multi-Cloud Scaffolding

TF • K8S • DOCKER

Scaffolds production Terraform, Kubernetes manifests, Docker Compose, and environment scripts targeting AWS, Azure, Google Cloud, or air-gapped on-prem.

Dedicated Green Leaf Git Hub

BITBUCKET • GITHUB • GITLAB

Integrated Git workflows: 1-click repo wizards, live branch diffs, AI-assisted commit message drafting, and PR creation without leaving the desktop app.

In-App Virtual Guide (IP Guardrails)

DOC INGESTION

Ingests client enterprise architecture documents, provides contextual workflow guidance, and delivers architectural deep-dives under strict IP guardrails.

DevOps Hygiene: Client deliverables land directly in their Git repos and cloud environments with clean commit histories.

ENTERPRISE ENGINEERING COMPARISON & PLATFORM SCORECARD

Enterprise Technical Scorecard: Status Quo vs. Evolve AI

The architectural, operational, and financial proof points that justify enterprise platform adoption.

ARCHITECTURAL DIMENSION	STATUS QUO AD-HOC CLOUD AI STACKS	EVOLVE AI ENTERPRISE STUDIO
Data Perimeter Security	Exfiltration risk to public LLM APIs over external HTTPS	**100% Air-Gapped Local Execution (Zero data leaves host)**
Determinism & Hallucination	Nondeterministic prompt guessing with high drift	**Mathematical Rule vs Model Gates (<5ms, 0% drift)**
Legacy Warehouse Integration	Read-only connectors or fragile custom ETL pipelines	**Native Oracle/DB2/Teradata with 2PC ACID write-back**
Infrastructure Governance	Manual, error-prone cloud setup across multiple consoles	**1-Click Terraform, Docker Compose & Kubernetes scaffolding**
Credential Management	Config files, dotfiles, or plaintext environment vars	**OS Keyring Native Vaults (Windows DPAPI, macOS Keychain)**
Hardware Utilization & TCO	Unpredictable per-token cloud bills & GPU cloud leases	**Runs on existing local CPUs/GPUs (Ollama, local Gemma)**

🏆 The VP of Engineering Advantage

Eliminates the maintenance nightmare of disconnected notebooks, brittle glue code, and shadow AI scripts. Delivers an **auditable, standardized engineering pipeline** that client teams can operate with confidence.

🚀 Immediate Evaluation & Pilot Deployment

Production Release v2.28.0 is packaged and ready for deployment. We onboard enterprise engineering teams through focused cohorts with direct core architect support: **support@evolveminds.com.au**.

Next Action: Schedule an architectural technical deep-dive or deploy the air-gapped evaluation binary today.